

Department of Geology

Sido Kanho Murmu University Dumka

Syllabus for FYUGP (Sem- I-VII)

As per the provision of NEP 2020

To be implemented from Academic Session 2023-27



Board of Studies: Department of Geology

1. Syed Raza Imam Rizvi
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Department of Geology & Prof in Charge
Sahibganj College, Sahibganj
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2. Dr Ranjit Kumar Singh
Assistant Professor
Department of Geology & Prof in Charge
Model College Rajmahal
3. Amber Xavier Kerketta
Assistant Professor & H.O.D.
Department of Geology, AS College, Deoghar

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External Member

4. Santosh Tudu
Assist Professor in Geology
Government Engineering College Banka

Signature *Signature* Amber Kerketta *Signature*

SEMESTER 1
GEOLOGY MJ Major 1
GEO-MJ-1: Earth System Science

Total Credits:4

FM:100

(End Sem + Internal Exam) 75+25=100 Marks

Unit 1: Earth as a planet:

Introduction to various branches of Earth Science; General characteristics and origin of the Universe, Solar System and its planets; Meteorites and Asteroids; Cosmic abundance of elements; Origin of Earth-atmosphere, ocean, and life.

Unit 2: Solid Earth:

Seismic waves and internal constitution of the Earth; Concept of isostasy; Earth's magnetic field; Geothermal gradient and internal heat of the Earth. Introduction to structure: Structural elements: planar and linear structures, concept of strike and dip, trend and plunge, rake/pitch

Unit 3: Plate Tectonics:

Concept of plate tectonics, sea-floor spreading and continental drift; Plate boundaries; Earthquake and earthquake belts; Volcanoes- types, products and their distribution.

Unit 4: Hydrosphere and Atmosphere:

Atmospheric circulations; Oceanic currents, tides and waves. Climate System and the Changing Climate from rock record; Concepts of eustasy.

Unit 5: Rock types and Soils:

Igneous, Sedimentary and Metamorphic rocks; Diastrophic and non-diastrorphic Structures; Application of primary sedimentary and igneous structures in structural geology. Concept of scale of observation of structures. Weathering and Erosion; Soil formation.

Unit 6: Understanding Stratigraphic records:

Stratigraphy and nature of stratigraphic records; Fundamental laws of stratigraphy: laws of superposition and faunal succession; Concepts of neptunism, plutonism, uniformitarianism, and catastrophism; Absolute and relative time in Geology. Unconformity and its types, recognition of unconformity. Concept of radiometric dating. Radiometric dating of rocks and minerals: U-Pb, Rb- Sr, Sm-Nd, C-14 methods. Geological time scale; Fossil record and Mass Extinction.

Unit 7: Natural Resources:

Mineral resources; hydrocarbon; Renewable energy resources.

SUGGESTED READINGS:

- 1 Brian J. Skinner, B. J. & Porter, S. C.: (2012). The Blue Planet: An Introduction to Earth System Science John Wiley & Sons. Inc.
- 2 Thompson G.R.R., Turk J. (1997) Introduction to Physical Geology. Brooks Cole.
- 3 Tarbuck, E. J. & Lutgens, F. K. (1998). Earth: An Introduction to Physical Geology. Pearson
- 4 Charles, C. P., Carlson, D., & Mcgeary, D. (2009) Physical Geology. McGraw-Hill Higher Education

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5 Duff, P. M. D., & Duff, D. (Eds.). (1993). Holmes' principles of physical geology. Taylor & Francis.

6 Emiliani, C. (1992). Planet earth: cosmology, geology, and the evolution of life and environment. Cambridge University Press.

Suggested Reference Books:

1 Grotzinger, J., Jordan, T.H., Press, F., Siever, R. (2007): Understanding Earth. W.H. Freeman & Co., New York, 5th Ed.

2 Skinner, B.J., Porter, S.C., Botkin, D.B. (1999): The Blue Planet – An Introduction to Earth System Science. John Wiley & Sons, Inc. New York. P.552.

3 Mathez, E.A. and Webster, J.D. (2004): The Earth machine – The Science of a Dynamic Planet. Columbia University Press, New York. P.335.

4 Gross, M. G. (1977). Oceanography: A view of the earth

SEMESTER 2

Geology MJ Major 2

GEO – MJ – 02 : Mineral Science

Credits:4

FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1:

Crystallography: Crystal--Concept of crystalline matter; Interfacial angle and external morphology in relation to internal structures; Crystal parameters and indices; form and zone.

Stereographic projection of crystal faces. Crystal symmetry, classification of crystals into systems and point groups. International symbol of point groups

Unit 2:

Atomic arrangements and Mineralogical structure: Atomic arrangements: Unit cell, CCP, FCC and HCP; Ionic radius and coordination, Pauling's rules. Solid Solution, Polymorphism, Pseudo morphism; Twinning.

Unit 3:

Rock forming minerals: Minerals - definition and classification, physical and chemical properties; Chemical classification of minerals; Internal structure, classification and

Composition of common rock-forming minerals (silicates); Derivation of structural formulae based on composition.

Unit 4:

Optical Mineralogy: Nature of light- Concept of visible electro-magnetic spectrum and optical behavior of minerals-- isotropic, uniaxial and bi-axial crystals; Double refraction; polarization, Nicol Prism; indicatrix; Introduction to petrological microscope; Refractive index and birefringence, interference phenomena, extinction, Michael Levy chart of interference colours, pleochroism, extinction. Interference phenomenon in convergent light, interference figures, and use of interference figures for determination of optic sign.

Suggested Reference Books:

1 Klein, C., Dutrow, B., Dwight, J., & Klein, C. (2007). The 23rd Edition of the Manual of Mineral Science (after James D. Dana). J. Wiley & Sons.

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- 2 Deer, W. A., Howie, R. A., & Zussman, J. (1992). An introduction to the rock-forming minerals (Vol. 696). London: Longman.
- 3 Nesse, W. D. (2011). Introduction to Optical Mineralogy (Fourth Edition). Oxford University Press.
- 4 Putnis, A. (1992): Introduction to Mineral Sciences. Cambridge University Press.
- 5 Whalstrom, E.E. (1969): Optical Crystallography. John Wiley & Sons
- 6 Verma, P. K. (2010). Optical Mineralogy (Four Colour). Ane Books Pvt Ltd.
- 7 Nesse, W.D., 2000, Introduction to Mineralogy, Oxford University Press, New York.

SEMESTER 2
Geology MJ Major 3
GEO – MJ – 03 :PRACTICAL
Credits:4
FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

Study of the symmetry of crystals. Stereographic projection of crystals.
Study of minerals in hand specimen - Silicates: olivine, garnet, andalusite, sillimanite, kyanite, staurolite, beryl, tourmaline, pyroxene, tremolite, hornblende-actinolite, serpentine, talc, muscovite, biotite, quartz, feldspar, nepheline, zeolite, asbestos, quartz
Other minerals: pyrite, chalcopyrite, galena, sphalerite, barite and gypsum, magnetite, haematite, pyrolusite, psilomelane, corundum, ilmenite, chromite, bauxite; fluorite, calcite, dolomite, apatite, graphite.
Introduction to optical microscope in laboratory studies.
Study of optical properties of common rock-forming minerals: quartz, orthoclase, microcline, plagioclase, perthite, nepheline, olivine, orthopyroxene, clinopyroxene, hornblende, staurolite, garnet, muscovite, biotite, calcite, tourmaline, sillimanite, kyanite, andalusite.
Study of major geomorphic features and their relationships with outcrops in topographic sheets.
True dip and apparent dip problems through graphical methods
Study of distribution of major stratigraphic units on the map of India.
Study of common sedimentary, igneous and metamorphic rocks in hand specimens – sandstone, shale, limestone, conglomerate, chert, BIF, granite, basalt, gabbro, dolerite, pegmatite, peridotite, anorthosite, nepheline syenite, rhyolite, slate, phyllite, schist, gneiss, and granulite

Suggested Reference Books:

- *Philpotts, A. and Ague, J. (2009). Principles of igneous and metamorphic petrology. Cambridge University Press.
- *Winter, J. D. (2014). Principles of igneous and metamorphic petrology. Pearson.
- *Raymond, L. A. (2002). Petrology: the study of igneous, sedimentary, and metamorphic rocks. McGraw-Hill Science Engineering.
- *Myron G. Best (2001). Igneous and Metamorphic Petrology.
- *Cox, K. G. and Bell, J. D. (1979). The Interpretation of Igneous Rocks. Springer/Chapman and Hall.
- *Bose M. K. (1997). Igneous Petrology.
- *Frost B. R. and Frost C. D (2014). Essentials of Igneous and Metamorphic Petrology. Cambridge University Press.
- *Pitcher, W.S. 1997. The Nature and Origin of Granitic Rock. Springer.
- *Wilson, M., 1989. Igneous Petrogenesis: a global tectonic approach, Cambridge University Press

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SEMESTER 3
Geology MJ Major 4
GEO – MJ – 04 :Structural Geology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Stress and Strain in Rocks

Concept of Stress: normal stress, shear stress, stress ellipse concept, principal axes of stress, planes of maximum shear stress, Mohr circle of stress.

Concept of strain: Longitudinal and shear strain, principal axes of strain, strain ellipse concept, Mohr circle for strain

Homogenous and inhomogeneous strain, Rotational and irrotational strain in rocks. Strain ellipsoids of different types and their geological significance. Flinn and Ramsay's diagram.

Basic methods of strain analysis

Rheological properties of rocks, Concept of rock deformation- brittle and ductile deformation, Factors controlling deformation behaviour of rocks.

Unit 2: Folds

Fold morphology and structural elements; Morphological classification of folds Outcrop patterns of folds, Geometric classification of folds

Mechanics of folding- buckling, bending. Kinematics of folding- flexural folding, flexural slip and flow folding, shear folding and passive folding

Superposed folding, morphological types, classification and basic geometric analysis in polydeformed terranes

Unit 3: Foliation and Lineation

Morphological features of foliations and lineations.

Tectonic significance of foliation and lineation, Relation of foliation and lineation with folds.

Brief idea of origin of foliation, Deformation mechanism, microstructure and fabric development

Unit 4: Faults, joints and shear zones

Classification of fractures- Faults and Joints

Joint- common terminology, characteristics and classification. Relation of Joints to Folds, exhumation and igneous bodies Fault zone terminology, Geometric classification of faults.

Effects of faulting on the outcrops, Criteria for recognition of faults, Fault zone rocks.

Anderson dynamic analysis of faulting, Characteristics of Normal, Thrust and Strike slip fault systems

Mechanics of fracturing and faulting, Fault plane solutions

Types of Shear zones and their kinematics, Shear zone rocks and shear sense indicators

Suggested Reference Books

- Davis, H.G, Reynolds, S.J, Kluth, C. F. (2011), Structural Geology of Rocks and Region, John Wiley
- Ragan, D. M. (2009) Structural Geology: an introduction to geometrical techniques (4th Ed.) Cambridge University Press (For Practical)
- Twiss, R. J. and Moores, E. M (2007) Structural Geology, Second Edition. W. H. Freeman and Company.
- Fossen, H (2010), Structural Geology, Cambridge University Press.

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- Marshak, S and Mitra G. (1988) Basic Methods in Structural Geology, Prentice Hall.
- Ben A. van der Pluijm and Stephen Marshak (2004) Earth Structure: An Introduction to Structural Geology and Tectonics (Second Edition) 2nd Edition
- Ghosh, S.K., 1993. Structural Geology: Fundamentals, and modern developments, Pergamon Press.
- Passchier, C. and Trouw, RAJ, 2005. Microtectonics. Springer, Berlin.
- Pollard, D.D. and Fletcher, R.C., 2005. Fundamentals of structural geology, Cambridge University Press.
- Ramsay, J.G and Huber, M.I., 1983. Techniques of Modern Structural Geology: Vol.I & II. Academic Press
- Ramsay, J. G, 1967. Folding and Fracturing of Rocks, McGraw-Hill Book Company, New York.
- Rowland, S.M., Duebendorfer, E. and Schiefelbein, I.M., 2007. Structural analysis and synthesis: a laboratory course in structural geology, Blackwell Pub.

SEMESTER 3
Geology MJ Major 5
GEO – MJ – 05 : PRACTICAL
Credits:4
FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

Topographic maps. Outcrop patterns of different structures. Stereographic projections of planes and lines

3-point problems, fold-fault problems and their solutions through graphical methods and stereographic projection methods.

Interpretation of geological maps with unconformity, fault, fold and igneous bodies. Construction of structural cross section.

Application of Borehole and Rotational Problems in Structural analyses

SEMESTER 4
Geology MJ Major 6
GEO – MJ – 06 : Geochemistry
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Basic Concepts: Introduction to properties of elements: Meteorite-classification and significance; Chemical bonding; Cosmic abundance of elements; Geochemical classification of elements.

Unit 2: Layered Structure of Earth and Geochemistry: Composition of the bulk silicate Earth; Composition of core; Composition of mantle: depleted mantle and enriched mantle; Composition of crust: Continental and Oceanic.

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Unit 3: Element transport: Advection and diffusion; Aqueous geochemistry- basic concepts and application in geological processes like Weathering, diagenesis & hydrothermal system; Eh, pH relation; Elements of marine chemistry; Geochemical behavior of elements.

Unit 4: Geochemistry of solid Earth: Geochemical behavior of elements during magmatic crystallization, partial melting; Concept of partition coefficient (Kd), compatible and incompatible elements; REE-essential characters, behavior and importance;

Unit 5: Isotope geology: Introduction to isotope geochemistry; Stable isotope geochemistry (C, O, S) and its application in geology; basic principles of Radioactive dating Methods: Rb- Sr, Sm-Nd, and C-14.

Suggested Reference Books:

- Mason, B. (1986). Principles of Geochemistry. 3rd Edition, Wiley, New York.
- Rollinson, H. (2007). Using geochemical data – evaluation, presentation and interpretation. 2nd Edition. Publisher Longman Scientific and Technical.
- Walther, J. V. (2009). Essentials of geochemistry. Jones and Bartlett Publishers.
- Albarède, F. (2003). Geochemistry: an introduction. Cambridge University Press.
- Faure, Gunter and Teresa M. Mensing (2004). Isotopes Principles and Applications. Wiley India Pvt. Ltd
- Killops S.D., Killops V.J. 2005. An introduction to organic geochemistry. 2nd ed. Blackwell Publishing, Malaysia.

SEMESTER 4

Geology MJ Major 7

GEO – MJ – 07 : Igneous Petrology

Credits:4

FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Introduction to Igneous Petrology: Magma generation in the crust and upper mantle. Physical properties of magma - temperature, viscosity, density and volatile content. Modes of emplacement of igneous rocks: volcanic, hypabyssal, plutonic.

Unit 2: Forms, Texture and Microstructure of Igneous rocks of Igneous rock bodies: Mode of occurrence of igneous rocks. Forms of igneous rocks. Crystallinity, granularity, shapes and mutual relations of grains; nucleation and growth of minerals in magma; Description of the following textures and microstructures with their occurrence in different rocks - panidiomorphic, hypidiomorphic, allotriomorphic, porphyritic, vitrophyric, poikilitic, ophitic, sub-ophitic, intergranular, intersertal, pilotaxitic, trachytic, graphic, granophyric, rapakivi, orbicular, corona, perthitic, myrmekitic, variolitic, speherulitic and spinifex.

Unit 3: Classification and Petrographical analysis of Igneous Rocks: Bases of classification of igneous rocks: mineralogical, textural, chemical, chemico-mineralogical and associational. Norm and mode. Standard classification schemes – Niggli, Hatch and Wells and IUGS. TAS diagram for volcanic rocks;

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Composition and texture of important igneous rocks: granitoids, pegmatite, syenite, monzonite, diorite, norite, gabbro, anthrothosite, dolerite, pyroxenites, peridotite, lamprophyres, carbonatite, rhyolite, andesite, dacite, basalt, komatiite.

Unit 4: Phase Diagrams: Phase rule and its application to eutectic, peritectic and solid solution system. Phase equilibria in the following binary and ternary systems under high dry and wet pressure with respect to their nature under low pressure (1 atmosphere), and their petrogenetic significance: diopside – anorthite, forsterite – silica, albite – anorthite, albite – orthoclase, diopside – albite – anorthite, forsterite – diopside – silica and nepheline - kalsilite – silica.

Unit 5: Diversification of igneous rock and Mantle Petrology: Bowen's reaction Series and its application, Magmatic differentiation- fractional crystallization, partial melting, assimilation and their role in magmatic differentiation. Bi-variate and tri-variate chemical variation diagram, idea about Mg-no., Fe-no., D.I. alkali-lime index, Petrgraphic Province. Mineralogy of upper mantle, phase transition in upper mantle, broad geochemical composition of upper mantle, variation in composition (mineralogical and geochemical) of upper mantle through space and time- mantle heterogeneity and its causes. Fluid phases in upper mantle.

Unit 6: Petrogenesis of Igneous Rocks: Petrogenesis felsic and mafic igneous rocks: granitoids (I-, S-, M-, and A-type), basalt, gabbros, anorthosite, alkaline rocks, kimberlites.

Unit 7: Magmatism in the following plate tectonic settings-

- Mid Oceanic Ridge setting
- Subduction Zone setting
- Continental and oceanic rift zone

Suggested Reference Books:

- Philpotts, A. and Ague, J. (2009). Principles of igneous and metamorphic petrology. Cambridge University Press.
- Winter, J. D. (2014). Principles of igneous and metamorphic petrology. Pearson.
- Raymond, L. A. (2002). Petrology: the study of igneous, sedimentary, and metamorphic rocks. McGraw-Hill Science Engineering.
- Myron G. Best (2001). Igneous and Metamorphic Petrology.
- Cox, K. G. and Bell. J. D. (1979). The Interpretation of Igneous Rocks. Springer/Chapman and Hall.
- Bose M. K. (1997). Igneous Petrology.
- Frost B. R. and Frost C. D (2014). Essentials of Igneous and Metamorphic Petrology. Cambridge University Press.
- Pitcher, W.S. 1997. The Nature and Origin of Granitic Rock. Springer.
- Wilson, M., 1989. Igneous Petrogenesis: a global tectonic approach, Cambridge University Press

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SEMESTER 4
Geology MJ Major 8
GEO – MJ – 08 : PRACTICAL

Credits:4

FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

Study of important igneous rocks in thin sections: granite, granodiorite, diorite, syenite, nepheline syenite, gabbro, anorthosite, ultramafic rock, basalt, andesite, dolerite, rhyolite, dacite, Lamprophyre and lamproite, pyroclastic rock and ignimbrite and alkaline rocks.

Norm calculation for silica undersaturated and silica oversaturated rocks

Plotting of modal data in IUGS classification diagram for plutonic rocks (Streckeisen diagram).

Numerical problems related to petrogenetic processes like fractional crystallization, partial melting, assimilation, magma mixing

Numerical/graphical problems on solid-liquid equilibrium system, magma viscosity, magma ascent rate

SEMESTER 5
Geology MJ Major 9
GEO – MJ – 09 : Sedimentary Petrology

Credits:4

FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Origin of sediments and Sediment granulometry: Physical and chemical weathering, soils and paleosols; Grain-size scale, particle size analysis and connotations; particle shape and fabric.

Unit 2: Sedimentary textures, structures and paleocurrent: Sediment transport mechanism- types of flow (Newtonian and Non-Newtonian), laminar and turbulent flow, subcritical, critical and supercritical flows; concept of mean flow velocity, unit discharge and bed shear stress; flow profile and flow separation; particle entrainment, transport and deposition, bedform stability diagram.

Sediment-gravity flow—types and deposits; Sedimentary structure- Primary, penecontemporaneous deformation and biogenic structures

Paleocurrent analysis: data acquisition, methodology, different palaeocurrent patterns

Unit 3: Sedimentary rocks: Components and classification(s) of conglomerates, sandstones, carbonate rocks and iron formations. General outline of controls on deposition of sandstones and carbonate rocks.

Unit 4: Diagenesis: Concepts of diagenesis, processes and stages of diagenesis, dolomites and dolomitisation

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Unit 5: Depositional environment: Facies models for fluvial, deltaic, siliciclastic shelf, carbonate platforms and deep-sea fan.

Unit 6: Tectonism and Sedimentation: Tectonic classification of sedimentary basins.

Unit 7: Stratigraphy: Principles of Litho-stratigraphy and Sequence Stratigraphy.

Suggested Readings:

- Allen, P.A., 1997. Earth Surface Processes, Blackwell publishing.
- Catuneanu, O., 2006. Principles of Sequence Stratigraphy. Elsevier, Amsterdam, 375 pp.
- Collinson, J.D. and Thompson, D.B., 1988. Sedimentary Structures, Unwin- Hyman, London.
- Folk, R.L. (1980) Petrology of Sedimentary Rocks. Hemphill Publishing Company, Austin, 184 p
- Leeder, M.R., 1982. Sedimentology: Process and Product. George Allen & Unwin, London, 344p.
- Lindholm, R.C., 1987. A Practical Approach to Sedimentology, Allen and Unwin, London.
- Pettijohn, F.J., 1975. Sedimentary Rocks, Harper and Row Publ. New Delhi.
- Miall, A.D., 1999. Principles of Sedimentary Basin Analysis 3rd Ed Springer Verlag, New York.
- Nichols, G., 1999. Sedimentology and Stratigraphy, Blackwell publishing.
- Reading, H. G., 1996. Sedimentary Environments: Processes, Facies and Stratigraphy, Blackwell Publishers
- Boggs, S., 1995. Principles of Sedimentology and Stratigraphy, Prentice Hall, New Jersey.
- Tucker, M.E., 2006. Sedimentary Petrology. Blackwell Publishing.
- Tucker, M.E. and Wright, V.P., 1990. Carbonate Sedimentology, Blackwell.
- Walker, R.G., Facies model 1976. Geoscience Canada.
- Prothero, D. R., & Schwab, F. (2004). Sedimentary geology. Macmillan.
- Posamentier, H.W., and Walker, R.G., 2006. Facies Models revisited, SEPM.

SEMESTER 5
Geology MJ Major 10
GEO – MJ – 10 : Metamorphic Petrology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Metamorphism: controls and types

- Definition of metamorphism
- Factors controlling metamorphism
- Types of metamorphism: contact, regional, fault zone metamorphism, impact metamorphism

Unit 2: Metamorphic Facies and Grades

- Index minerals, metamorphic zones and isograds.
- Concept of metamorphic facies and grade
- Composition-paragenesis diagrams. ACF, AKF and AFM diagrams
- Schreinemakers' analyses, petrogenetic grid and phase equilibria modelling
- Concept of metamorphic P-T-t path

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Unit 3: Metamorphism and deformation

- Structure and textures of metamorphic rocks
- Relationship between metamorphism and deformation.

Unit 4: Metamorphic reactions

- Types of metamorphic reactions
- Kinetics of metamorphic reactions
- Progressive metamorphism of pelitic, basic and carbonate rocks

Unit 5: Quantification of metamorphic processes

- Geothermobarometry
- Ultrahigh temperature (UHT) and ultrahigh pressure (UHP) metamorphism
- Metamorphic fluid and metasomatism
- Geochronology of metamorphic rocks

Unit 6: Metamorphism and crustal evolution

- Anatexis and migmatites
- Partial melting of crustal rocks

Unit 7: Tectonics and metamorphism

- Heat flow and thermal models of orogenic belts
- Metamorphism along convergent and divergent plate margin
- Metamorphic styles through ages

Suggested Reference Books :

- Philpotts, A., & Ague, J. (2009). Principles of igneous and metamorphic petrology. Cambridge University Press.
- Winter, J. D. (2014). Principles of igneous and metamorphic petrology. Pearson.
- Rollinson, H. R. (2014). Using geochemical data: evaluation, presentation, interpretation. Routledge.
- Yardley, B. W. D. (1989). An introduction to metamorphic petrology. Longman Scientific and Technical, London.
- Spear F. S. 1993. Metamorphic phase equilibria and Pressure-Temperature-Time paths. Mineralogical Society of America Monograph 799.

SEMESTER 5
Geology MJ Major 11
GEO – MJ – 11 :PRACTICAL

Credits:4
FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

Exercises on sedimentary structures in hand specimens; Particle size distribution & statistical treatment,

Palaeocurrent analysis, Petrography of clastic and non-clastic rocks through thin sections and their environmental inference.

Textural and mineralogical study of metamorphic rocks in thin sections: varieties of schists, amphibolite, charnockite, khondalite, mafic granulite.

Graphical plots of metamorphic mineral assemblages using chemographic diagrams

Calculation of P and T using geothermobarometry

Construction P-T grid using Schreinemakers' rules

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SEMESTER 6
Geology MJ Major 12
GEO – MJ – 12 : Stratigraphy
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Introduction to Precambrian Stratigraphy

Divisions of Precambrian time scale, Characteristics and status of Archaean and Proterozoic Eons in global perspective, Archaean-Proterozoic boundary.
Models of Precambrian tectonics and crustal development

Unit 2: Physiographic and Tectonic subdivisions of India

Brief Introduction to the physiographic and tectonic subdivisions of India.
Study of geological map of India and identification of major Precambrian stratigraphic units.
Introduction to Indian cratons, mobile belts and Proterozoic sedimentary basins

Unit 3: Geologic evolution of important Precambrian terrains in India

Geologic evolution with emphasis on sedimentation, lithology, magmatism, structure, metamorphism and geochronology of: Singhbhum Craton, Dharwar Craton, Bastar Craton, Aravalli-Bundelkhan Craton; Central Indian Tectonic Zone, Chhotanagpur Granite Gneiss Belt, Eastern Ghats Belt, Southern Granulite Terrane; Vindhyan and Cudappah basins

Unit 4: Introduction- Definition; Distribution of major Phanerozoic basins in India; Important stratigraphic boundaries during Phanerozoic time in India - a. Precambrian-Cambrian boundary, b. Permian-Triassic boundary, and c. Cretaceous-Tertiary boundary.

Unit 5: Important Indian successions- Detailed study of important Phanerozoic successions in India with emphasis on lithology, structural and tectonic set up as well as paleo-environment of the following:

- Gondwana succession of Peninsular India;
- Successions of extra-peninsular India, with special reference to Kashmir and Spiti valley;
- Mesozoic successions in peninsular India with special reference to Kutch and Cauvery basins;
- Cenozoic successions: Siwalik succession, Assam and Bengal basins.

Unit 6: Stratigraphy of the Deccan Traps and Intertrappean beds.

Unit 7: Quaternary Geology- Definition; Principles of subdivisions; Major paleoclimatic and paleo-geographic events in Quaternary period with special emphasis on the Indian Subcontinent; Quaternary successions in India.

Unit 8: Evolution of life: Indian context- Indian fauna and flora in the light of evolution as well as global and regional geological phenomena: Gondwana flora and vertebrates; Major invertebrate groups from Kashmir, Spiti, Kutch and Cauvery; Siwalik vertebrates; Human evolution in Indian context.

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Suggested Reference Books:

- Krishnan, M.S., 2006, Geology of India and Burma, 6th Edn., CBS, 536 p. [Earlier editions also useful]
- Ramakrishnan, M. and Vaidyanathan, R., 2008, Geology of India, Vol. 1, Geological Society of India, Bangalore.
- Pascoe, E.H., 1950, A manual of the geology of India and Burma (3 volumes), 3rd Edn., Geological Survey of India.
- Valdiya K.S. (2010). The making of India, Macmillan India Pvt. Ltd.

SEMESTER 6
Geology MJ Major 13
GEO – MJ – 13 : Palaeontology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Taxonomy and Species concept: Species concept with special reference to palaeontology; Taxonomic hierarchy; Binomial nomenclature

Unit 2: Invertebrate Palaeontology: Brief introduction to important invertebrate groups (e.g., Trilobita, Mollusca); their biostratigraphic and evolutionary significance

Unit 3: Micropalaeontology: Brief introduction to important microfossil groups (e.g., Foraminiferida); their biostratigraphic and evolutionary significance; introduction to palynology

Unit 4: Introduction to Vertebrate Palaeontology: Origin and divisions of vertebrates; Major turning points in vertebrate evolution – evolution of jaw, terrestrialization, amniote evolution, evolution of flight, evolution of mammals.

Unit 5: Introduction to Palaeobotany: Origin and divisions of plants; Major steps in plant evolution.

Unit 6: Organic skeleton as a machine – An introduction to the relation between morphology and life habit; mechanical properties of benthic, oceanic (planktonic, nektonic) and terrestrial (flying, land-dwelling) organisms; concept of allometry in ontogeny.

Suggested Reference Books:

- Foote, M. and Miller, I.A. (2007) Principles of Paleontology. 3rd Edition by W. H. Freeman and company
- Clarkson, E. N. K. (2012) Invertebrate paleontology and evolution. 4th Edition by Blackwell Publishing.
- Benton, M. (2009). Vertebrate paleontology. 4th Edition by John Wiley & Sons.
- Arnold, C.A. (2018) An Introduction to Paleobotany. Surjeet Publications
- Shukla, A. C., & Misra, S. P. (1975). Essentials of paleobotany. Vikas Publisher
- Armstrong, H. A., & Brasier, M.D. (2005) Microfossils. Blackwell Publishing.

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SEMESTER 6
Geology MJ Major 14
GEO – MJ – 14 : Environmental Geology & Mathematical Geology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Environmental Geology

Unit 1:

Definition and dimensions of environment; General idea about components and composition of different environmental domains such as atmosphere, hydrosphere and biosphere; Types of environmental pollution;

Unit 2:

Introduction to weather and climate; Past-climates in the earth history; Concept and origin of monsoon; Elements of natural hazards.

Unit 3:

Definition and concept of environmental geology. Processes of soil formation, types of soil, soil degradation and mitigation.

Unit 4:

Environmental changes due to the influence of geological events. volcanoes, earthquakes, floods and anthropogenic activities. Environmental degradation due to mining and related activities and remedies. Water and air pollution.

Mathematical Geology

Unit 5:

Application of statistics, trigonometry, algebra and calculus in Geology

Books Recommended:

1. Valdiya, K. S., (1987) Environmental Geology - Indian Context. Tata McGraw Hill New Delhi.
2. Keller, E. A., (2000) Environmental Geology. Shales E. Merrill Publishing Co., Columbus, Ohio.
3. Montgomery, C., (1984) Environmental Geology. John Wiley and Sons, London.
4. Sharma J. P., Environmental Studies, Laxmi Publications (P) Ltd, New Delhi
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SEMESTER 6
Geology MJ Major 15
GEO – MJ – 15 : PRACTICAL
Credits:4
FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

- Unit 1:** Mapping of major stratigraphic units from India.
- Unit 2:** Studying major features of paleogeographic maps from Phanerozoic.
- Unit 3:** Study of diagnostic morphological characters, systematic position, stratigraphic position and age of various invertebrate (including microfossils), vertebrate and plant fossils from India.
- Unit 4:** Exercises on lithostratigraphic, biostratigraphic and chronostratigraphic correlation.
- Unit 5:** Study of fossils showing various modes of preservation
- Unit 6:** Functional morphology of invertebrates (including microfossils)
- Unit 7:** Plant morphology and climate
- Unit 8:** Vertebrate teeth and feeding habit
- Unit 9:** Water & air pollution Case studies.

SEMESTER 7
Geology MJ Major 16
GEO – MJ – 16 : Economic Geology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

- Unit 1: Introduction to ore geology:**
Basic concepts of Ores, gangue minerals, tenor, grade, resources and reserves. Morphology of ore bodies. Review of concepts of ore genesis process
- Unit 2: Ore-forming processes**
Orthomagmatic processes, Sedimentary processes, Hydrothermal
Supergene and Residual enrichment processes & Metamorphic processes
- Unit 3: Plate Tectonics and ore deposits**
Role of plate tectonics in ore mineralization. Metallogeny through ages
- Unit 4: Nuclear Fuel**
Important minerals.
Types of U and Th deposits. U and Th deposits in India
- Unit 5: Metallic and Non-metallic ores of India**
Metallic ores
Non-metallic and industrial rocks and minerals Atomic minerals
Gem & Gemstones
- Unit 6: Fossil Fuel**
Elementary idea regarding origin, uses and distribution of coal and petroleum in India.

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Suggested Reference Books:

1. Evans, A.M. (1993) Ore Geology and Industrial minerals. Wiley
2. Laurence Robb. (2005) Introduction to ore forming processes. Wiley.
3. Ridley, J. (2013): Ore Deposit Geology. Cambridge University Press, UK. P398.
4. Guilbert, J.M. and Park Jr., C.F. (1986) The Geology of Ore deposits. Freeman & Co.
5. Bateman, A.M. and Jensen, M.L. (1990) Economic Mineral Deposits. John Wiley.
6. Gokhale, K.V.G.K. and Rao, T.C. (1978) Ore deposits of India their distribution and processing, Tata-McGraw Hill, New Delhi.
7. Deb, S. (1980) Industrial minerals and rocks of India. Allied Publishers.
8. Sarkar, S.C. and Gupta, A. (2014) Crustal Evolution and Metallogeny in India. Cambridge Publications.
9. Mukherjee, A. (1999): Ore Genesis – A Holistic Approach. Allied Publishers Ltd., New Delhi, India. P657.

SEMESTER 7
Geology MJ Major 17
GEO – MJ – 17 : Engineering Geology
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1:

Engineering properties of rocks, laboratory testing of rocks – uniaxial compressive strength, tensile strength, shear strength, Mohr's diagram: mass movement- classification and causes".

Unit 2:

Dam – its type, geotechnical consideration, selection of sites, forces acting on a dam.

Unit 3:

Reservoirs – geological investigation, silting of reservoirs.

Unit 4:

Tunnel – geological investigations in tunnelling. Case histories related to Indian civil Engineering projects.

Suggested Reference Books:

1. Blyth, F.G.H. and M. H. de Freitas(1984)Geology for Engineers, Butterworth- Heinemann Title
2. Krynine, D.P and Judd, W.R (2005) Principles of Engineering Geology and Geotechniques, CBS Publishers & Distributors
3. Ries, H. and T. L. Watson, (1949) Elements of Engineering Geology, New York, JohnWiley & Sons, Inc.

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4. Krynine, D.P. and Judd W.R. 1957. Principles of Engineering Geology and Geotechnique, McGraw Hill.
5. Johnson, R.B. and De Graf, J.V. 1988. Principles of Engineering Geology, John Wiley
6. Goodman, R.E., 1993. Engineering Geology: Rock in engineering constructions. John Wiley & Sons, N. Y.
7. Engineering and General Geology - Parbin Singh.
8. Bell: F.G-, 2006. Basic Environmental and Engineering Geology Whittles Publishing.

SEMESTER 7
Geology MJ Major 18
GEO – MJ – 18 : Remote Sensing & GIS
Credits:4
FM:100

Theory (End Sem + Internal Exam) 75+25= 100 Marks

Unit 1: Fundamentals of remote sensing, Electromagnetic Radiation (EMR), Satellite, sensors and scanners, data formats; Remote sensing techniques: Optical, thermal and hyperspectral remote sensing.

Unit 2: Digital Image Processing, Image Errors, Rectification and Restoration, FCC, Image Enhancement, Filtering, Image Rationing, Image classification and accuracy assessment. GIS integration and Case studies-Indian Examples

Unit 3: Hyperspectral remote sensing - Basic concept, spectroscopy, use of spectroradiometer, spectral signature library, hyperspectral sensors and imagery. Application of hyperspectral remote sensing in geological studies.

Unit 4: Active remote sensing: Synthetic Aperture Radar (SAR) and Side Looking Airborne Radar (SLAR) – principle and applications in geological studies.

Unit: 5: Remote sensing application in lithological mapping, structural mapping, mineral exploration, groundwater exploration. Introduction to DEM analysis.

Unit-6: GIS: Spatial and non-spatial data analysis; applications in prospecting natural resources, risk assessment of natural hazards. GNSS and GPS: Recent development; Applications in earth system sciences.

Suggested Reference Books:

1. Demers, M.N. (1997). Fundamentals of Geographic Information System, John Wiley and sons. Inc.
2. Hoffmann-Wellenhof, B., Lichtenegger, H. and Collins, J. (2001). GPS: Theory and Practice, Springer Wien, New York.
3. Jensen, J.R. (1996). Introductory Digital Image Processing: A Remote Sensing Perspective. Springer- Verlag.
4. Lillesand, T. M. and Kiefer, R.W. (2007). Remote Sensing and Image Interpretation. Wiley.
5. Richards, J.A. and Jia, X. (1999). Remote Sensing Digital Image Analysis. Springer-Verlag.

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SEMESTER 7
Geology MJ Major 19
GEO – MJ – 19 :PRACTICAL

Credits:4

FM:100

Practical (End Sem + Internal Exam) 75+25= 100 Marks

1. Recognition of Dam site through geological map.
2. Ore identification in hand specimen.
3. Ore identification under the microscope
4. Plotting of important ore deposits in map of India
5. Reserve calculation of ore
6. Problems related to photogrammetry
7. Practical record and viva-voce

SEMESTER 1

MN Minor from Discipline:1 Geology

GEO-MN-01: Essentials of Geology

To be opted by other Major Students

Credits: 4 (Theory: 3 + Practical: 1)

FM: 100 (75 + 25)

Theory (End Sem + Internal Exam) 60+15=75 Marks

Unit 1: Introduction to geology:

Scope, sub-disciplines and relationship with other branches of sciences.

Unit 2: Earth in the solar system:

Origin. Earth's size, shape, mass, density, rotational and evolutionary parameters.

Solar System- Introduction to various planets - Terrestrial Planets, Jovian Planets.

Unit 3: Solid Earth, Hydrosphere, Atmosphere and Biosphere

Mechanical layering of the Earth: lithosphere, asthenosphere, mantle and core.

Earthquake and earthquake belts: seismic waves and internal constitution of the Earth.

Volcanoes and volcanism, distribution of volcanoes.

Concept of isostasy.

Formation of core, mantle, crust, atmosphere, hydrosphere and biosphere.

Convection in Earth's core and production of its magnetic field.

Geothermal gradient and internal heat of the Earth.

Unit: 4. Rocks, Mineral and fossils

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Definition. General character. Usefulness.

Unit 5: Plate Tectonics

Fundamental Earth process: plate tectonics.

Plates and plate boundaries.

Origin of oceans, continents, mountains and rift valleys.

Unit 6: Earth's Surface Processes

Weathering and Erosion. Landforms in deserts, glaciated region and river valleys.

Unit 7: Age of the earth:

Radioactivity and its application in determining the age of the Earth.

Suggested Reference Books:

Holmes' Principles of Physical Geology. (1992). Chapman and Hall.

Emiliani, C, (1992). Planet Earth, Cosmology, Geology and the Evolution of Life and Environment. Cambridge University Press.

Gross, M.G. (1977). Oceanography: A view of the Earth. Prentice Hall.

Practical: 25 Marks

GEO – MN – 01- LAB

Study of minerals in hand specimen - Silicates: talc, muscovite, biotite, quartz, feldspar, asbestos, quartz

Ore minerals: pyrite, chalcopryite, galena, sphalerite, barite and gypsum, magnetite, haematite, pyrolusite, psilomelane.

Study of major geomorphic features and their relationships with outcrops in topographic sheets.

Study of distribution of major stratigraphic units on the map of India.

Study of common sedimentary, igneous and metamorphic rocks in hand specimens

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SEMESTER 3
MN Minor from Discipline: Geology
GEO-MN-02: Crystallography & Mineralogy
To be opted by other Major Students
Credits: 4 (Theory: 3 + Practical: 1)
FM: 100 (75 + 25)
Theory (End Sem + Internal Exam) 60+15=75 Marks

Unit 1: Crystallography

Elementary ideas about crystal structure; Crystal: faces, edges, solid angles and interfacial angles; Crystallographic axes and axial angles; Elements of Crystal Symmetry. Parameter, indices and symbol. Laws of crystallography; Isometric System.

Unit 2: Optical Mineralogy

Nature of light; Polarization of light; Isotropic and anisotropic substances (minerals); Phenomenon of double refraction; Nicol prism – its construction; Construction and function of petrological microscope. Optical properties of Quartz, orthoclase, plagioclase, microcline, biotite, muscovite.

Unit 3: Descriptive Mineralogy

Mineral - definition; Physical properties of minerals; Structure and classification of silicates; Chemical composition and diagnostic physical properties of the following rock forming minerals - quartz, orthoclase, microcline, kyanite, calcite, gypsum, apatite, talc, fluorite, garnet, biotite, muscovite, beryl.

Suggested Reference Books:

1. Elements of crystallography and mineralogy Wade & Mattox.
2. Phillips, F.C. (1971): Introduction to Crystallography, Longman Group Publ.
3. Rutleys Elements of Mineralogy, CBS Publication.

Practical: 25 Marks

GEO – MN – 02- LAB

1. Crystal drawing of the following forms- cube, Octahedron, Dodecahedron
2. Identification of rock forming minerals in hand specimen.
3. Identification of rock forming minerals under microscope
4. Record and viva-voce

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SEMESTER 5

MN Minor from Discipline: Geology

GEO-MN-03: Structural Geology & Petrology

To be opted by other Major Students

Credits: 4 (Theory: 3 + Practical: 1)

FM: 100 (75 + 25)

Theory (End Sem + Internal Exam) 60+15=75 Marks

Structural Geology

Unit 1:

Understanding dip and strike. Construction and uses of Clinometer and Brunton compass; Elementary idea and nomenclature of Folds, Faults, Unconformity and Joints.

Petrology

Unit 2:

Igneous Rocks - Definition and classification of igneous rocks; Forms of igneous bodies; Brief petrographic description of common igneous rocks such as – granite, gabbro, dolerite, basalt, rhyolite

Unit 3:

Sedimentary rocks – texture and structure; Brief Petrographic descriptions of common sedimentary rocks such as – conglomerate, breccia, sandstone, limestone and shale.

Unit 4:

Metamorphic rocks – definition, types and its agents; Textures of metamorphic rocks; Brief Petrographic descriptions of common metamorphic rocks – slate, phyllite, schist, gneiss, marble and quartzite.

Suggested Reference Books:

1. Davis, G. R. (1984) Structural Geology of Rocks and Region. John Wiley
2. Billings, M. P. (1987) Structural Geology, 4th edition, Prentice-Hall.
3. Park, R. G. (2004) Foundations of Structural Geology. Chapman & Hall.
4. Petrology – G. W. Tyrell
5. Petrology – Ehlers and Blatt
6. Sedimentary petrology – Tucker M. E.

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Practical: 25 Marks
GEO – MN – 03- LAB

1. Study of simple geological maps from No 1 to 8 involving simple Dip, Fold, Fault and Unconformity, drawing of geological sections and detailed geological description.
2. Identification of igneous, sedimentary and metamorphic rocks in hand specimen.
3. Record and viva-voce

SEMESTER 7
MN Minor from Discipline: Geology
GEO-MN-04: Stratigraphy & Paleontology
To be opted by other Major Students
Credits: 4 (Theory: 3 + Practical: 1)
FM: 100 (75 + 25)
Theory (End Sem + Internal Exam) 60+15=75 Marks

Unit 1: Stratigraphy

Stratigraphy: Principles of stratigraphy. Geological time scale. A brief account of the important geological formation of India viz. Precambrian of Jharkhand, Gondwana Supergroup of Jharkhand and Vindhyan Supergroup.

Unit 2: Palaeontology

Palaeontology: Definition and its scope; Fossils definition, their mode of preservation and uses; Morphology and geological history of the following: Brachiopoda, Palecypoda and Trilobita.

Unit 3: Hydrogeology

Occurrence and vertical distribution of groundwater; Hydrological cycle; Water bearing properties of rocks - Porosity and permeability, Specific yield, specific retention and storage coefficient. Aquifers and their types.

Suggested Reference Books:

1. Kumar, R. (1985): Historical Geology and Stratigraphy of India, Wiley Eastern Ltd.
2. Shrock, R.R. and Twenhoffel, W.H. (1952): Principles of Invertebrate Paleontology, CBS Publ.
3. Woods, H. (1985): Palaeontology Invertebrate, CBS Publ.
4. Todd, D. K. (1995): Groundwater hydrology, John Wiley and Sons.

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Practical: 25 Marks

GEO – MN – 04 – LAB

1. Identification of important invertebrate and plant fossils
2. Record and viva-voce.

SEMESTER 1/2/3
Multidisciplinary MDC
GEO – MDC: Introduction to Geology
Credits - 3: (Theory)
FM: 75

Unit 1 Introduction to geology:

Scope, sub-disciplines and relationship with other branches of sciences.

Unit 2 Earth in the solar system,

Origin. Earth's size, shape, mass, density, rotational and evolutionary parameters. Solar System- Introduction to various planets - Terrestrial Planets, Jovian Planets.

Unit 3 Solid Earth, Hydrosphere, Atmosphere and Biosphere.

Volcanoes and volcanism, distribution of volcanoes.

Concept of isostasy.

Formation of core, mantle, crust, atmosphere, hydrosphere and biosphere.

Unit: 4. Rocks, Mineral and fossils.

Definition. General character. Usefulness.

Unit 5 Plate Tectonics

Fundamental Earth process: plate tectonics. Plates and plate boundaries. Origin of oceans, continents, mountains and rift valleys.

Unit 6 Earth's Surface Processes

Weathering and Erosion. Landforms in deserts, glaciated region and river valleys.

Suggested Reference Books:

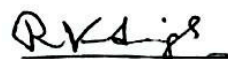
Holmes' Principles of Physical Geology. (1992). Chapman and Hall.

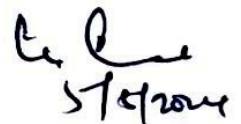
Emiliani, C, (1992). Planet Earth, Cosmology, Geology and the Evolution of Life and Environment. Cambridge University Press.

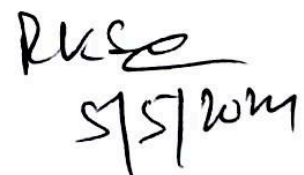
Gross, M.G. (1977). Oceanography: A view of the Earth. Prentice Hall.

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